Cole Koester

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Summary: Software engineer with 7 years of experience building and maintaining production distributed systems. Enjoys working across the stack and at scale.

Professional Experience

Lyft Inc. — New York, NY

Software Engineer, Production Infrastructure & Marketplace Forecasting — 11/2021 - Present

Forecasting Platform: Scaled marketplace forecasting to support a 50x increase in data throughput.

- Delivered two flagship forecasting models powering marketplace pricing at finer granularity (50x), serving 7 features from a single model where each prior model served one.
- Architected and implemented regional partitioning, splitting geohash regions into homogeneous subregions; improved relative MAPE by 15% and cut forecast message size and memory usage by 80%.
- Unblocked the scale-up by implementing compression across every API surface of the platform, reducing data transfer size by 10x with marginal increase in CPU utilization.
- Spearheaded a new S3 pointer sinking pattern enabling customers to consume features too large for the API layer, designed to integrate cleanly alongside existing sink patterns.
- Drove team planning and roadmap definition, including an async training data loading initiative projected to reduce memory usage by 20GB per pod.

Zone Aware Routing: Improved intra availability zone routing with ROI of $2 million.

- Reduced inter-AZ production traffic by 40%, saving $2M annually on data transfer costs.
- Migrated 1,416 microservices serving 500K+ requests/second to Envoy load balancing subsets.
- Deprecated error prone load balancing components in favor of configuring load balancing subsets in Envoy.
- Wrote design spec and Grafana/Kibana dashboards. Communicated with customer teams to debug load balancing edge cases.

No More Yaml (NoMoYa): Decreased time-to-deploy networking settings from 15 minutes to less than 30 seconds.

- Implemented new configuration API server (Go) and user interface (TypeScript + React) handling circuit breakers, health check endpoints, traffic migrations, and network dependency allow lists.
- Improved team operations through self-service SEV mitigation, preventing context switching from team members.

Control Plane Backend Sharding: Collaborated with tech lead to simplify endpoint discovery.

- Transitioned from leader-elected writers to independent writers, enhancing service reliability and simplifying the deployment pipeline.
- Implemented new data layer on control plane frontend and backend with 0 downtime, and a 99.95% mesh availability SLA.
- Parallelized service discovery queue reducing endpoint query latency by roughly 50%.

Operations & Leadership

- Participated in debugging and mitigating more than 100 incidents through analyzing various Kibana and Prometheus queries, and SSHing into hosts themselves to validate networking components.
- Performed technical deep dives on load balancing, did 30+ candidate interviews, actively participated in team planning, and acted as mentor for both interns and new-hires.
- Part of interview revamp working group. Developing new interview standards to address changes in the interview space related to LLMs.

iManage LLC — Chicago, IL

Software Engineer, API — 11/2019 - 08/2021

IP Whitelisting and Subdomains: Implemented requirements with ROI of $1.8 M / Year.

- Implemented new authorization service in Go scaling to roughly 10M requests / day. Authorization service handled: vanity subdomains, IP whitelisting, and proxying to core auth service.

API Gateway: Deployed new API gateway edge proxy serving 30M requests / day with a 99.95% success rate SLA.

- Defined and deployed an Envoy edge proxy configuring separate listeners for internal/external clients, and multi-region traffic.
- Deployed rate-limit service with a centralized configuration repo, reducing DDoS risk and putting backpressure on noisy internal services.

Education

New York University — Courant Institute of Mathematical Sciences — New York, NY

Graduate Coursework in Scientific Computing — Numerical Methods I, Partial Differential Equations — 2026 - Present

Tufts University — Somerville, MA

Bachelor of Science in Mathematics | QuestBridge Scholar

Skills

- **Languages**: Go, TypeScript, Python, Bash, Java, C++, C, Rust
- **Tools**: Envoy, AWS, Kubernetes, Terraform, Docker, React, Prometheus/PromQL, Grafana, Kibana, TensorFlow, Xarray, Pandas